

2011000105030001
EXAMINATION JULY 2024 (PURAK EXAM)
BACHELOR OF COMPUTER APPLICATIONS
(FIFTH SEMESTER)
NETWORK TECHNOLOGIES (503)

[Time: As Per Schedule]

[Max. Marks : 70]

Instructions:

1. Fill up strictly the following details on your answer book
 - a) Name of the Examination: **BACHELOR OF COMPUTER APPLICATIONS (FIFTH SEMESTER)**
 - b) Name of the Subject: **NETWORK TECHNOLOGIES (503)**
 - c) Subject Code No: **2011000105030001**
2. Sketch neat and labelled diagram wherever necessary.
3. Figures to the right indicate full marks of the question.
4. All questions are compulsory.

Seat No:

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Student's Signature

Q.1 Select an appropriate option.

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1. Topology requires a multipoint connection.
 - a) Star
 - b) Mesh
 - c) Ring
 - d) Bus
2. Application layer is implemented in _____.
 - a) End system
 - b) NIC
 - c) Ethernet
 - d) Packet transport
3. Which of the following is false with respect to UDP?
 - a) Connection-oriented
 - b) Unreliable
 - c) Transport layer protocol
 - d) Low overhead
4. The functionalities of the presentation layer include _____.
 - a) Data compression
 - b) Data encryption
 - c) Data description
 - d) All of the mentioned
5. Communication channel is shared by all the machines on the network in _____.
 - a) Broadcast network
 - b) Unicast network
 - c) Multicast network
 - d) Anycast network

6. Fiber optics posses following properties _____
 - a) Immune electromagnetic interference
 - b) Very less signal attenuation
 - c) Very hard to tap
 - d) All of the mentioned
7. Which is not an application layer protocol?
 - a) HTTP
 - b) SMTP
 - c) FTP
 - d) TCP
8. To deliver a message to the correct application program running on a host, the _____ address must be consulted.
 - a) IP
 - b) MAC
 - c) Port
 - d) None of the mentioned
9. Electronic mail uses which Application layer protocol?
 - a) SMTP
 - b) HTTP
 - c) FTP
 - d) SIP
10. ICMP is primarily used for _____.
 - a) Error and diagnostic functions
 - b) Addressing
 - c) Forwarding
 - d) Routing

Q.2 Answer the following (Any Four)

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1. Explain client-server architecture.
2. Explain intranet. What is the difference between internet and intranet? Explain its architecture.
3. What is MANET? Explain its types.
4. Explain SMTP, DNS, and POP protocol in details.
5. Define URL. Explain types of URL.
6. Explain different types of Networks.

Q.3 Answer the following (Any Four)

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1. Explain SSL, HTTP, FTP, Telnet protocol in details.
2. List out topology. Explain any 2 topologies in details.
3. What is search Engine? Explain the process of locating website.
4. Explain Network Devices: Hub, switch and Routers.
5. Explain concept of Email and its working and services.
6. Explain Guided Transmission Media.

Q.4 Write short Note on (Any Two)

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1. Difference between http and https.
2. Explain TCP header.
3. Explain Term: Unicast, Multicast and Broadcast.
